Biconical antenna with inhomogeneous dielectric lens for UWB applications

A.M. Bobreshov, A.S. Zhabin✉, E.A. Seregina and G.K. Uskov

The improvement of the ultra-wideband (UWB) biconical antenna directional characteristics at the upper part of the bandwidth by filling the aperture with the inhomogeneous dielectric lens is discussed in this Letter. The inhomogeneity of the filling was achieved by the complex three-dimensional configuration of the air gaps and the dielectric with constant permittivity. The distribution law of permittivity in the aperture was obtained in the geometrical optics approximation using the law of Lihtenekker for finely dispersed composition of substances with different dielectric constants. The numerical simulations and the experimental measurements of the biconical antenna characteristics have been carried out and the results are presented. The application of the proposed method allows to reduce the level of lateral radiation and to increase the antenna gain and bandwidth. The technique for the lens manufacturing by using 3D printing technologies is presented.

Introduction: Biconical antennas are widely used as parts of ultra-wideband (UWB) radio systems when an omnidirectional radiation pattern in the azimuthal plane is required [1, 2]. Their bandwidth and gain in the main direction of radiation depend on the antenna geometry, which is illustrated in Fig. 1. The lower limit of the bandwidth is determined by the height of the cones (h) and the upper one is limited by the stability of the radiation pattern [1]. The distortions in the upper part of the bandwidth take place due to the phase incursion of the electromagnetic field ΔP in the aperture (between points A and C in Fig. 1). Another important characteristic of the antenna is the input impedance. It depends on the angle α between the generatrix line of the cone and the axis OX (Fig. 1). The optimal radiation and matching characteristics are achieved with α in the range from 30° to 40° [3, 4]. Thus, the antenna design must fulfil three conditions for effective radiation and good matching characteristics in a wide frequency band:

(i) the maximum wavelength of the radiated signal

$$\lambda_{\max} = 2h \cdot (\varepsilon_{\text{eff}})^{1/2}, \tag{1}$$

where ε_{eff} is the effective dielectric constant of the material in the space between the cones of the antenna;
(ii) the optimal matching and radiation characteristics are achieved with α in the range from 30° to 40°;
(iii) the minimum wavelength of the radiated signal is determined by the phase incursion $\Delta P<\pi$.

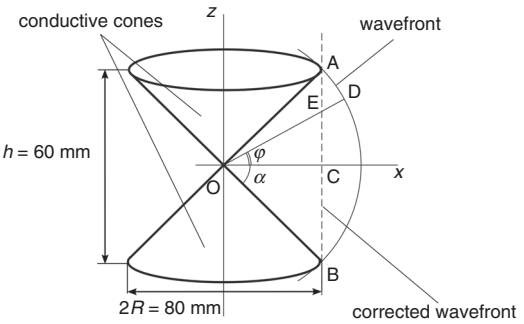


Fig. 1 *Beam path in the antenna aperture*

Thus, the lowest frequency of the operating range of the biconical antenna depends on its geometrical dimensions (h, R and α) and these dimensions limit the maximum radiated frequency due to the phase incursion in the antenna aperture. Therefore, an additional technique should be applied to expend the operating frequency range and improve the antenna performance at the upper part of the bandwidth. One of the methods to reduce the phase incursion in the aperture is the use of lens from metamaterials [5].

This Letter is focused on reducing the distortions of the radiation pattern by introducing the inhomogeneous dielectric structure into the space between the conductive cones of the biconical antenna. The permittivity of this filling varies with a direction of electromagnetic wave propagation. The distribution law of permittivity in the space was obtained in the geometrical optics approximation, which states that electromagnetic waves propagate from a single phase centre according to the laws of geometric optics. As well, the Lihtenekker law for finely dispersed compositions and foamed materials was used to derive the distribution law of permittivity within the space between cones of the antenna. The technique for manufacturing the lens with a complex configuration of the dielectric material and air gaps using three-dimensional printing technologies is presented in this Letter. The results of simulations and experiments confirm that the proposed method allows to reduce the phase difference in the aperture and improve the directional characteristics of the biconical antenna.

Method for compensating phase incursion in antenna aperture: The electromagnetic field in the aperture is described by the Maxwell's equations with boundary conditions for the antenna geometry. Solving such equations in the general form is a very complicated task, therefore the approximation of geometric optics was used. The fundamental assumptions of geometric optics are a straight-line propagation of an electromagnetic wave and the existence of a point source of radiation – a phase centre. As the antenna has axial symmetry, carrying out the cross-sectional analysis is sufficient. Fig. 1 shows a beam path in the antenna aperture. Straight lines OA and OB are generatrix lines of the cones, OC is the radius of the cone, point O is the hypothetical phase centre. The time of the beam propagation from point O to any point of the line AB should be the same so as to get rid of the phase shift of the wavefront on the line. Thus, all the beams emanating from the phase centre at an arbitrary angle should reach the aperture in the same period of time: $t_{\text{OC}} = t_{\text{OE}} = t_{\text{OA}}$ or $t_\alpha = t_\varphi$. Taking into account the dependency of the electromagnetic wave propagation velocity on the electrical properties of the medium, the law of permittivity distribution in the antenna aperture can be obtained

$$\varepsilon(\varphi) = \varepsilon_{\max} \cos^2(\varphi), \tag{2}$$

where $\varepsilon_{\max}$ is the maximum value of the permittivity of the dielectric material, which is selected to meet the conditions of physical implementation [6]: $\varepsilon(\varphi)>1$.

Thus, the necessary delay of a beam propagation is introduced by filling the antenna aperture with the medium, whose distribution law of permittivity is in accordance with (2). The permittivity increases with decreasing the angle between a beam and the axis x. So, the maximum delay is produced in the direction $\varphi = 0$, and the minimum delay – in the direction $\varphi = \alpha$.

Simulation of biconical antenna with inhomogeneous dielectric lens: The distribution law of permittivity in the antenna aperture (2) is a smooth function of φ. Today, it is not possible to produce a material whose properties can be described with such function. The dependency of the permittivity on the angle was approximated by a stepped-index function (Fig. 2). According to the stepwise approximation, the space between the cones was divided into subareas. The subareas were bounded above and below by cones with different heights and the same radius of a basis. The permittivity of each subarea was constant, as is illustrated on the graph in Fig. 2.

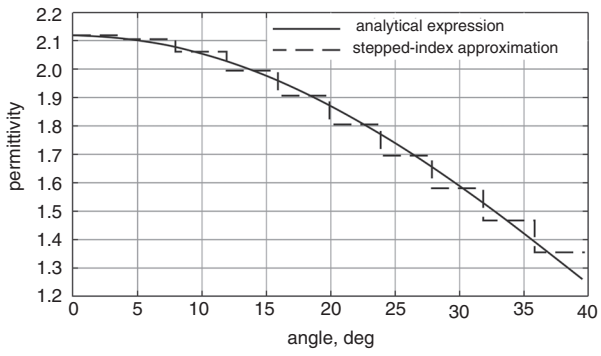


Fig. 2 *Dependency of the dielectric permittivity on the angle*

The biconical antenna with and without the dielectric lens has been simulated using software CST Microwave Studio. The permittivity of

the lens varied with the angle φ as shown in Fig. 2. Radiation patterns obtained by simulation are presented in Fig. 3. As can be seen from the graphs, with increasing frequency side lobes appear (Fig. 3: 8 GHz) in the radiation pattern and the gain in the main direction decreases (Fig. 3: 14 GHz) for the antenna without the dielectric lens. The radiation maximum does not coincide with the main direction at frequencies higher than a certain value (Fig. 3: 18 GHz). The use of the proposed dielectric lens allows to reduce the level of the side lobes and to focus the main beam in the plane perpendicular to the axis of symmetry (axis z in Fig. 1).

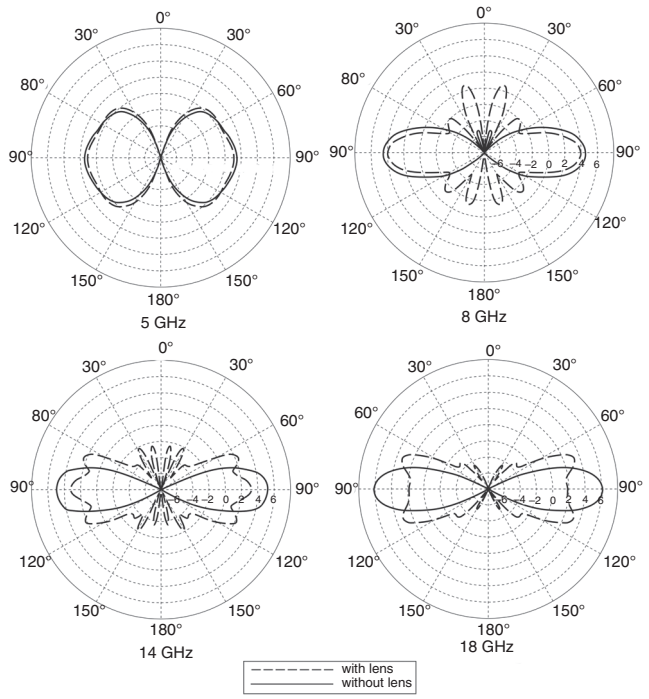


Fig. 3 *Radiation pattern of the biconical antenna at the frequency of 5, 8, 14 and 18 GHz*

Synthesis of structure with variable permittivity: The method of manufacturing a filling with a variable permittivity using a perforated dielectric has been proposed in [7]. A perforated dielectric is considered as a disordered finely dispersed composition of a dielectric and air. Thus, such material should obey the Lihtenekker law for finely dispersed composition of substances with different permittivity. This law in the case of air gaps is

$$\varepsilon_{\text{eq}} = \varepsilon_p^{\theta p}, \tag{3}$$

where ε_{eq} is the equivalent permittivity of a cell, ε_p is the permittivity of dielectric, θ_p is a cell filling ratio, which can be calculated as $V_P/(V_P + V_{\text{air}})$. V_P is a volume of a cell occupied by dielectric, V_{air} is a volume of a cell occupied by air.

The structure of the lens for the biconical antenna is presented in Fig. 4 in two projections: in the longitudinal section (Fig. 4*a*) and in the cross-section (Fig. 4*b*). In the process of producing the prototype of the proposed lens, the space inside the antenna was divided into layers with a thickness of 0.4 mm. Each layer consisted of a set of cells, which are parts of a sector bounded by two circles with different radii. The difference in the radii of the neighbouring circles remained constant and was equal *dR*. The value *a* for each cell is the difference between radii of circles, which limit an air gap in each sector (Fig. 4*a*). The filling ratio of a cell determined its equivalent permittivity. Thus, the necessary ratio from the formula (3) can be obtained for any cell by calculating the permittivity in its geometric centre according to (2). The prototype of the biconical antenna with the dielectric lens using the fused deposition modelling (FDM) technology with the MakerBot Replicator 2X 3D printer is presented in Fig. 4c. High impact polystyrene material was used as a dielectric due to the low losses in the microwave range [7].

Measuring biconical antenna characteristics with inhomogeneous dielectric lens: The experimental measuring of the biconical antenna characteristics has been carried out using the Agilent PNA-L 5230C vector network analyser. The voltage standing wave ratio (VSWR) and the gain in the main direction of radiation were used as main characteristics to estimate the antenna performance. As shown from the graphs in Figs. 5 and 6, the experimental measurements have a good agreement with the results of the simulation.

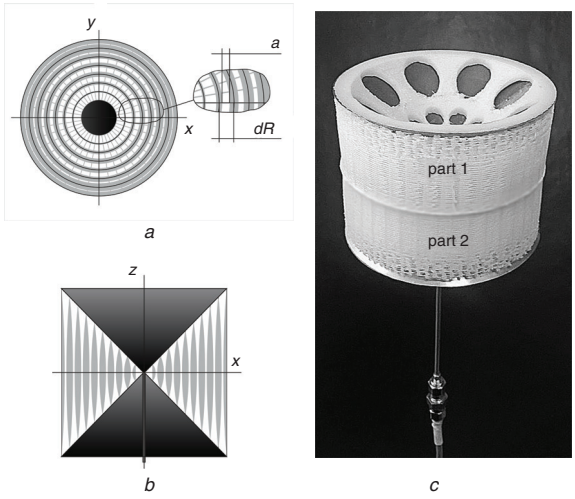


Fig. 4 *Structure and the prototype of the dielectric lens for the biconical antenna*

a Longitudinal section at an arbitrary height
b Cross-section in the centre of the antenna
c Prototype of the antenna with the dielectric lens

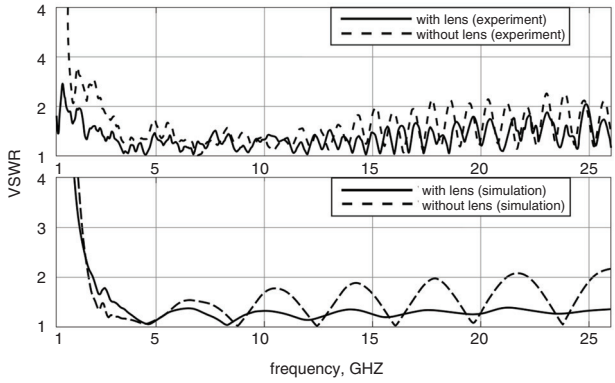


Fig. 5 *VSWR of the biconical antenna obtained by simulation and experiments*



Fig. 6 *Gain of the biconical antenna obtained by simulation and experiments*

The dielectric constant of the medium in the antenna aperture increases with the lens application. Therefore, the maximum wavelength of the radiated signal increases according to the formula (1). Thus, the

introduction of the lens allows not only to improve the radiation pattern, but also to reduce the lowest frequency of the operating range by 1.5 GHz at VSWR<2 (Fig. 5). The difference between the calculated and measured VSWR in Fig. 5 can be explained by the presence of the air-coaxial connector of the device under test, which was not taken into account in the numerical simulation.

The dependence of the gain in the main direction of radiation on frequency is shown in Fig. 6. As can be seen from the graph, the dielectric lens application in the experiments provides the increase of the gain varying from 1 to 5.5 dB in the frequency band from 8 to 26 GHz. The minor difference (less than 1 dB) between calculated and experimentally measured dependences of the gain of the biconical antenna is explained by inevitable inaccuracies in the lens manufacturing by FDM printing technology due to temperature compression and the material expansion.

Designs of the biconical antenna from [1–4] were analysed. The values of the main parameters of these designs are shown in Table 1. F_{min} and F_{max} are the minimum and maximum frequencies of the matching operating range (VSWR<2). G_{max} is the maximum gain of radiation in the main direction. Due to the phase difference in the antenna aperture, the radiation pattern distorts in the upper part of the bandwidth and the gain in the main direction decreases. The distortions of the radiation pattern in [1–4] also rise with increasing frequency. The magnitude of the side lobes at the frequency of 6 GHz is less than 4 dB lower than this of the main lobe in [1]. The distortions of the radiation pattern take place at frequencies above 6.85 GHz in [2] and above 11 GHz in [3]. The application of the proposed dielectric lens for the biconical antenna shifts the frequency, at which these distortions appear, higher than the upper limit of the matching operating range (VSWR<2).

Table 1: Summary of biconical antenna parameters

Reference	R, mm	h, mm	F_{min}, GHz	F_{max}, GHz	G_{max}, dB
[1]	29	54	1.5	41	4.7
[2]	29	63.3	3.1	10.6	5
[3]	43	50	2	12	3.2
[4]	6.2	19.25	3.7	4.5	2.7
this Letter without lens	40	60	3	18	4.8
this Letter with lens	40	60	1.5	26	8

Conclusions: This Letter is devoted to the method for improving the performance of the biconical antenna by introducing the inhomogeneous dielectric structure in the aperture. The dielectric lens improves both the directional properties of the antenna and the matching characteristics. The application of the proposed method in the experiments allows to increase the gain by 1–5.5 dB in the frequency band from 8 to 26 GHz up to 8 dB. The lowest frequency of the operating range was reduced by 1.5 GHz at the level of VSWR<2. This Letter also describes the technique for manufacturing the proposed dielectric lens and demonstrates that it can be produced by using three-dimensional printing technologies. The characteristics of the biconical antenna with and without the inhomogeneous dielectric lens obtained experimentally are in good agreement with the simulation results.

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